

Quantum Bayesian Networks: Compositionality and Typing via Linear Logic

Rémi Di Guardia, Thomas Ehrhard, Claudia Faggian

IRIF – Inria, CNRS, Université Paris Cité

Proof Society 2026, 10 September 2026



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Quantum – Quick & Dirty

Quantum for this talk:

- **Data/Object** =

state in a *Hilbert Space* $\mathcal{H}$ respecting some properties

$$|0\rangle \langle 0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

- **Operation/Morphism** =

linear maps between Hilbert Spaces preserving the properties

$$\mathcal{E} : \rho \mapsto |0\rangle \langle 0| \rho |0\rangle \langle 0|$$

- **Tensor product** $\otimes$

- **Measure** from a Hilbert Space to its dimension $\mathcal{H} \rightarrow X$,
probabilistic & modify the input state

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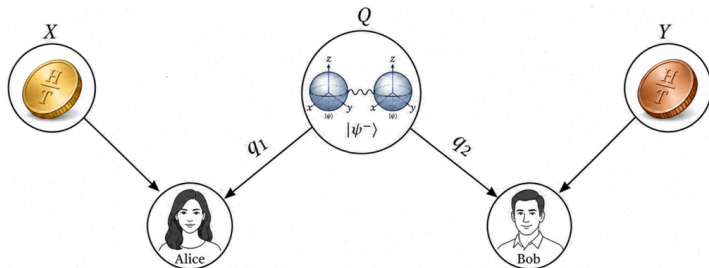
$$\mathcal{E} : \rho \mapsto |0\rangle \langle 0| \rho |0\rangle \langle 0|$$

- **Tensor product** $\otimes$
- **Measure** from a Hilbert Space to its dimension $\mathcal{H} \rightarrow X$,
probabilistic & modify the input state

Differences with classical:

- **No cloning:** states cannot be cloned/duplicated/broadcast
- **Entanglement:** given states in $\mathcal{H}_1 \otimes \mathcal{H}_2$, measuring the state in $\mathcal{H}_1$ may modify the state in $\mathcal{H}_2$

Quantum – Bell's Experiment



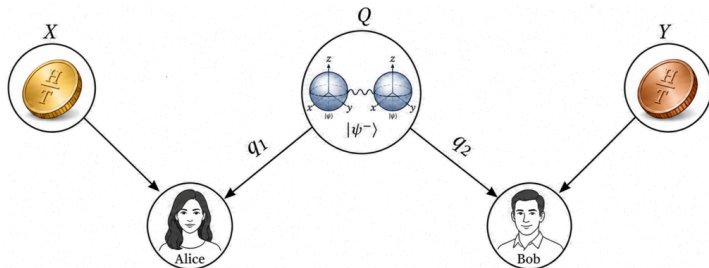
Q = Preparation of the entangled q_1 and q_2

X = Flip a coin

Alice = Measure on q_1 parameterized by the value of X

Question: $\Pr(\text{Alice} = 0, \text{Bob} = 0 \mid X = 1, Y = 0) = ?$

Quantum – Bell's Experiment



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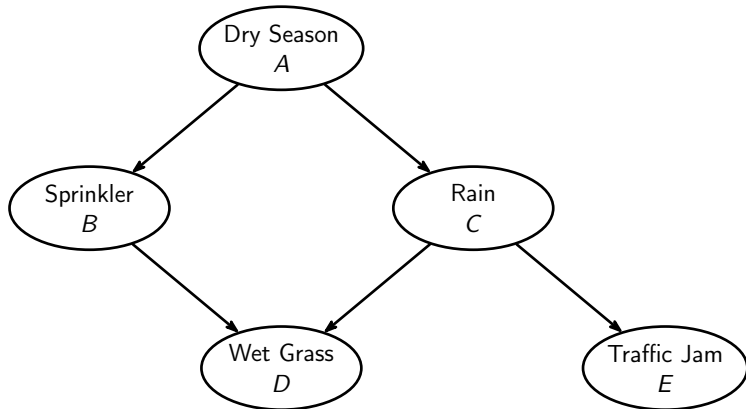
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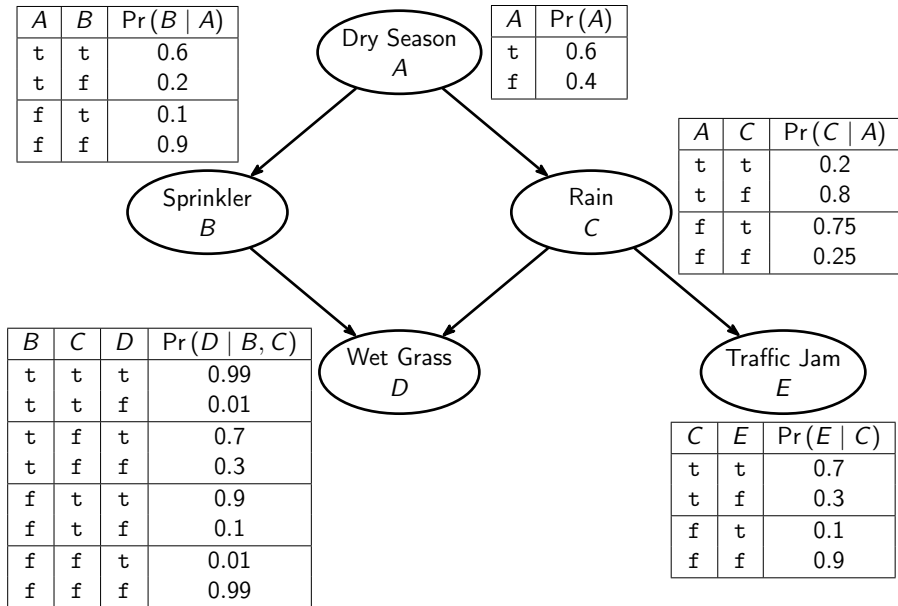
Question: $\Pr(\text{Alice} = 0, \text{Bob} = 0 \mid X = 1, Y = 0) = ?$

Quantum theory as a theory of inference: Given a quantum system, what is the *probability* of obtaining some measurement?

Bayesian Networks



Bayesian Networks



Bayesian Networks

A	B	$\Pr(B A)$
t	t	0.6
t	f	0.2
f	t	0.1
f	f	0.9



A	$\Pr(A)$
t	0.6
f	0.4

A	C	$\Pr(C A)$
t	t	0.2
t	f	0.8
f	t	0.5
f	f	0.25

EFFICIENT INFERENCE ALGORITHMS

B	C	D	$\Pr(D B, C)$
t	t	t	0.3
t	t	f	0.7
t	f	t	0.9
t	f	f	0.1
f	t	t	0.01
f	t	f	0.99
f	f	t	0.01
f	f	f	0.99

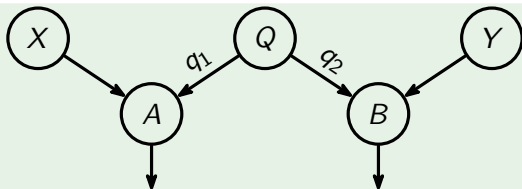
GRAPHICAL CAUSAL MODEL

Traffic Jam		
E	C	$\Pr(E C)$
t	t	0.7
t	f	0.3
f	t	0.1
f	f	0.9

Quantum Bayesian Networks

Definition: Quantum Bayesian Networks [Henson, Lal, Pusey 2014]

DAG with a family of **quantum operations** per node such that composing these top-down yields a *probability distribution*.



$$Q : \mathbb{C} \rightarrow \mathcal{H}_1 \otimes \mathcal{H}_2$$

= Preparation of the entangled q_1 and q_2

$$X : \text{a map } f_x : \mathbb{C} \rightarrow \mathbb{C} \text{ for each value } x \text{ of } X$$

(with $\sum_x f_x(1) = 1$)

= Flip a coin

$$A : \text{a map } g_{x,a} : \mathcal{H}_1 \rightarrow \mathbb{C} \text{ for each value } x \text{ of } X \text{ and } a \text{ of } A$$

(with $\sum_a g_{x,a}(q_1) = 1$)

= Measure on q_1 parameterized by the value of X

Quantum Bayesian Networks

Definition: Quantum Bayesian Networks [Henson, Lal, Pusey 2014]

DAG with a family of **quantum operations** per node such that composing these top-down yields a *probability distribution*.

Limits

Compositionality: given a decomposition, is the data of the *whole* obtainable from the data of the *parts*?

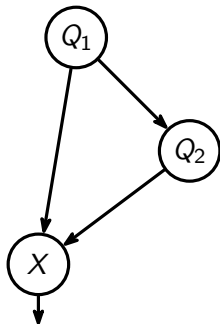
(graphs good as more decompositions, advantage of BN)

Modularity: do two *parts* compose into a *DAG*?

Contributions

Solutions through **another presentation**
of Quantum Bayesian Networks

Limits – Compositionality



$$\phi^{Q_1} : \mathbb{C} \rightarrow \mathcal{H}_1 \otimes \mathcal{H}_2$$

$$\phi^{Q_2} : \mathcal{H}_2 \rightarrow \mathcal{H}_3$$

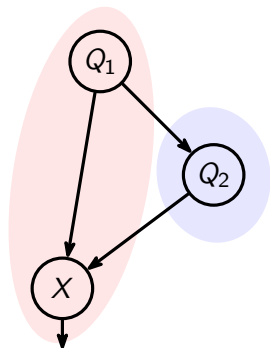
$$\phi^X(x) : \mathcal{H}_1 \otimes \mathcal{H}_3 \rightarrow \mathbb{C}$$

Full network:

$$\phi^{Q_1, Q_2, X}(x) : \mathbb{C} \rightarrow \mathbb{C}$$

$$= \phi^X(x) \circ (id_{\mathcal{H}_1} \otimes \phi^{Q_2}) \circ \phi^{Q_1}$$

Limits – Compositionality



$$\phi^{Q_1} : \mathbb{C} \rightarrow \mathcal{H}_1 \otimes \mathcal{H}_2$$

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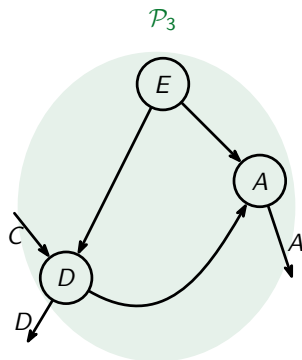
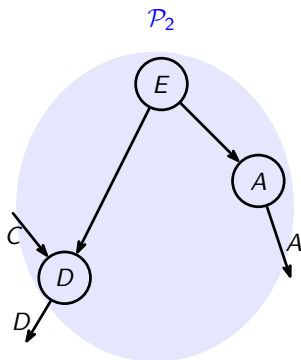
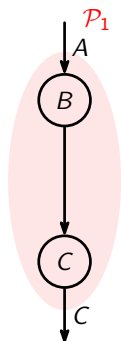
$$\phi^{Q_1, X}(x) : \mathcal{H}_3 \rightarrow \mathcal{H}_2$$

$$\phi^{Q_2} : \mathcal{H}_2 \rightarrow \mathcal{H}_3$$

$\Rightarrow \phi^{Q_1, Q_2, X}(x)$ is not the **composition** of $\phi^{Q_1, X}(x)$ and ϕ^{Q_2} !
(typed either $\mathcal{H}_3 \rightarrow \mathcal{H}_3$ or $\mathcal{H}_2 \rightarrow \mathcal{H}_2$)

Idea: Functions do not work well at graph level, **matrices** would be better
(also in Bayesian networks: factors instead of conditional probabilities)

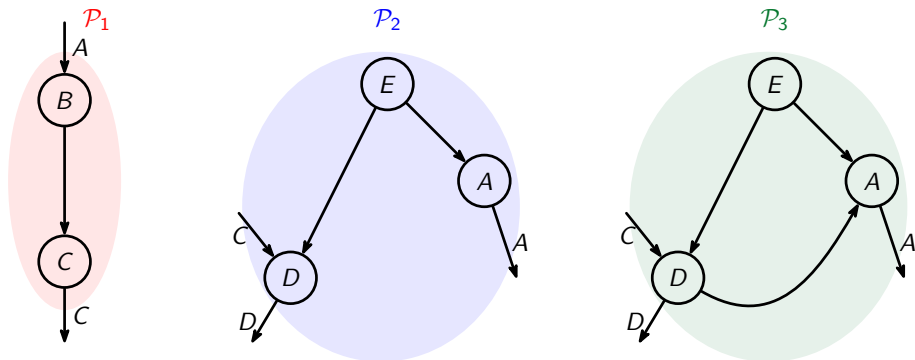
Limits – Modularity



- Part P_1 waits for input A and outputs C
- Parts P_2 and P_3 wait for input C and output A and D

Is it “legal” to branch P_1 to P_2 ? P_1 to P_3 ?

Limits – Modularity



- Part P_1 waits for input A and outputs C
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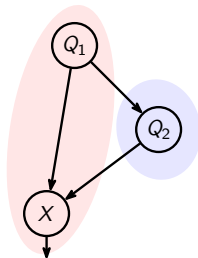
Is it “legal” to branch P_1 to P_2 ? P_1 to P_3 ?

→ $P_1 \cup P_2$ is a QBN but $P_1 \cup P_3$ is not (cycle $B \rightarrow C \rightarrow D \rightarrow A \rightarrow B$)

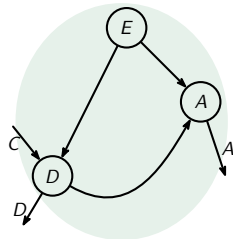
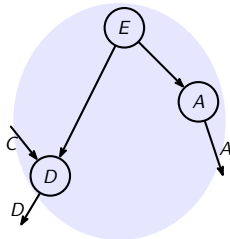
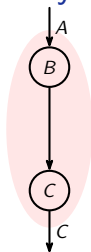
Idea: Inputs & Outputs are insufficient, we need a **type**
Instead of a new syntax use **proof-nets** = typed graphs from Linear Logic

Plan

► Compositionality by Quantum Factors



► Modularity by Typing via Linear Logic

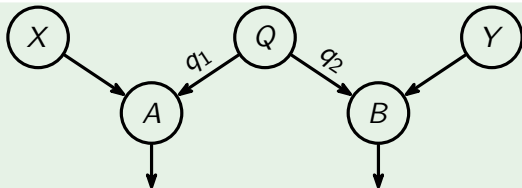


Quantum Factors

Not a *quantum operation* per node but a:

Definition: Quantum Factor

Let $(X_1, \dots, X_n)$ be random variables and $(\mathcal{H}_1, \dots, \mathcal{H}_m)$ Hilbert spaces. A **Quantum Factor** on $(X_1, \dots, X_n, \mathcal{H}_1, \dots, \mathcal{H}_m)$ is a function ϕ from $\prod_{i=1}^n \text{Val}(X_i)$ to positive matrices in $\bigotimes_{j=1}^m \mathcal{H}_j$.



$$Q : \{\star\} \rightarrow \mathcal{H}_1 \otimes \mathcal{H}_2$$

= Preparation of the entangled q_1 and q_2

$$X : \text{Val}(X) \rightarrow \mathbb{C}$$

= Flip a coin

$$A : \text{Val}(X) \times \text{Val}(A) \rightarrow \mathcal{H}_1$$

= Measure on q_1 parameterized by the value of X

Quantum Extension of a Classical Concept

Classical Factor over rvs X

$$\phi : Val(X) \rightarrow \mathbb{R}_0^+$$

Quantum Factor over rvs X and quantum $\mathcal{H}$

$$\phi : Val(X) \rightarrow \mathcal{L}(\mathcal{H})^+$$

Quantum Extension of a Classical Concept

Classical Factor over rvs X	Quantum Factor over rvs X and quantum $\mathcal{H}$
$\phi : Val(X) \rightarrow \mathbb{R}_0^+$	$\phi : Val(X) \rightarrow \mathcal{L}(\mathcal{H})^+$

Quantum Factors come with a **product** $\odot$ and a **sum** Σ

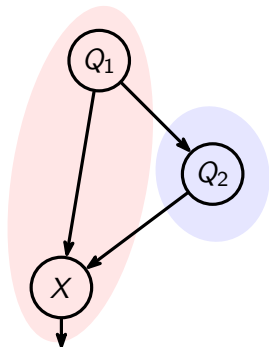
	Pure Classical	Pure Quantum
Theory	factors of Bayesian Networks	$\otimes$ -networks
$\odot$	product of factors	contraction
Σ	marginalization	partial trace
!	efficiency by sharing	no-cloning theorem

Technical Challenges:

- **sharing** for classical vs **no-cloning** for quantum
- **conservative** extension of (classical) Bayesian Networks

(Equivalent to the original definition of QBN by the process/state duality.)

Compositionality by Quantum Factors



$$\phi^{Q_1} : \mathbb{C} \rightarrow \mathcal{H}_1 \otimes \mathcal{H}_2$$

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Full network:

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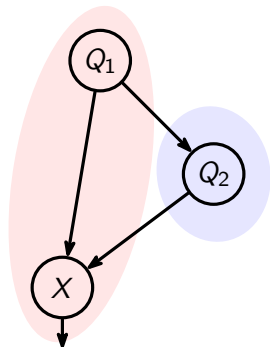
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Compositionality by Quantum Factors



$$\phi^{Q_1} : \mathcal{H}_1 \otimes \mathcal{H}_2$$

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Full network:

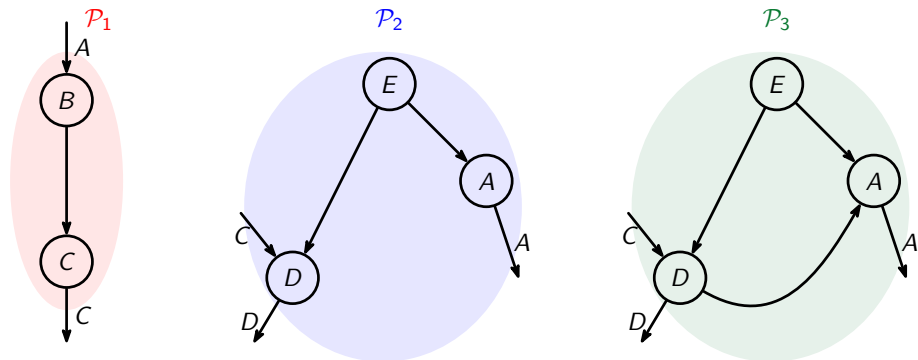
$$\begin{aligned}\phi^{Q_1, Q_2, X} : \text{Val}(X) &\rightarrow \mathbb{C} \\ &= \phi^X \odot \phi^{Q_2} \odot \phi^{Q_1}\end{aligned}$$

$$\begin{aligned}\phi^{Q_1, X} : \text{Val}(X) &\rightarrow \mathcal{H}_3 \otimes \mathcal{H}_2 \\ \implies \phi^{Q_1, Q_2, X} &= \phi^{Q_1, X} \odot \phi^{Q_2}\end{aligned}$$

$$\phi^{Q_2} : \mathcal{H}_2 \otimes \mathcal{H}_3$$

More generally, can compute for **any order** of the nodes!

Modularity



Observation: $\mathcal{P}_1 \cup \mathcal{P}_2$ is a QBN but not $\mathcal{P}_1 \cup \mathcal{P}_3$

How to ensure two parts always form a QBN?

Idea: We add a **type** = an interface

Could do so by patching QBN and getting yet another new syntax...

We prefer to use **proof-nets** = typed graphs from Linear Logic

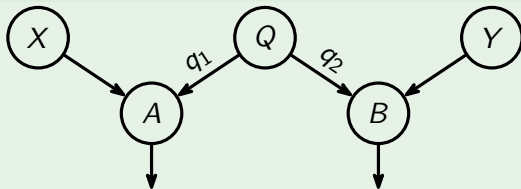
Proof-Nets for Quantum Bayesian Networks

Definition: Proof-Net

Graph respecting some **graphical criterion** and built from:

$$\begin{array}{c}
 X^- \vdash^{ax} X^+ \quad q^- \vdash^{ax} q^+ \quad A^\perp \vdash_{cut} A \quad A \vdash_{\otimes} B \quad A \vdash_{\multimap} B \quad X^- \vdash_c X^- \quad w \\
 |A \otimes B \quad |A^\perp \multimap B \quad |X^- \quad |X^-/q^- \quad \boxed{X/\otimes q_i} \\
 \begin{array}{|c|} \hline \vdots \\ \hline Y_1^- \\ \hline \end{array} \quad \begin{array}{|c|} \hline \vdots \\ \hline q_1^- \\ \hline \end{array} \quad X^+/\otimes q_i^+
 \end{array}$$

Example



Exact Characterization by Proof-Nets

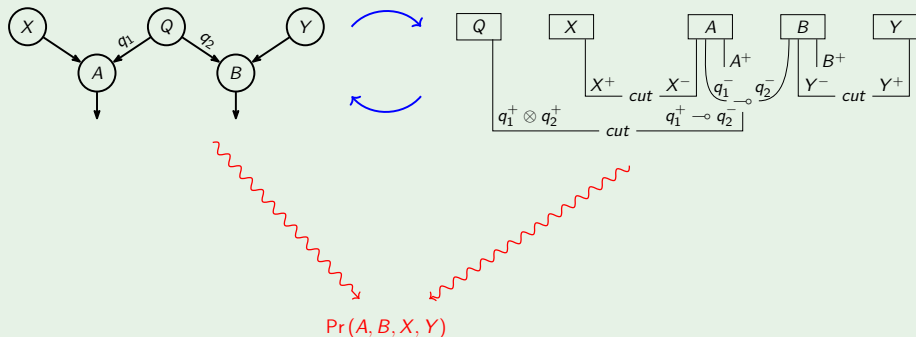
Theorem

There is a **one-to-one correspondence** between Quantum Bayesian Networks and (a class of) Proof-Nets.

that preserves the semantics

= the associated probability distribution

Example



Exact Characterization by Proof-Nets

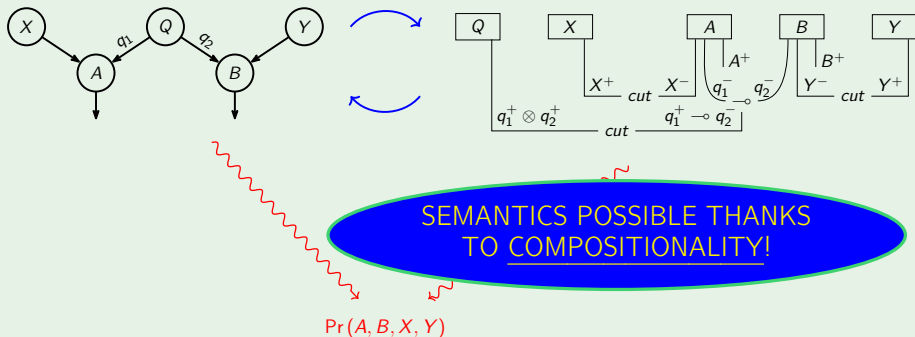
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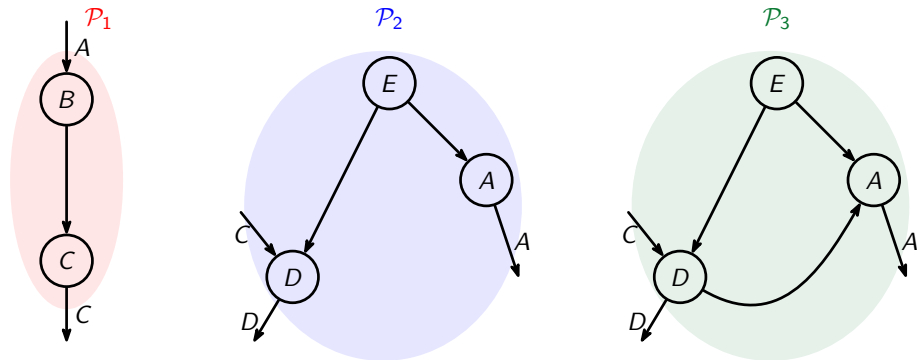
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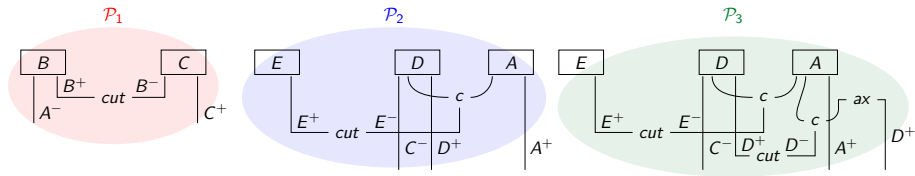


Modularity by Typing



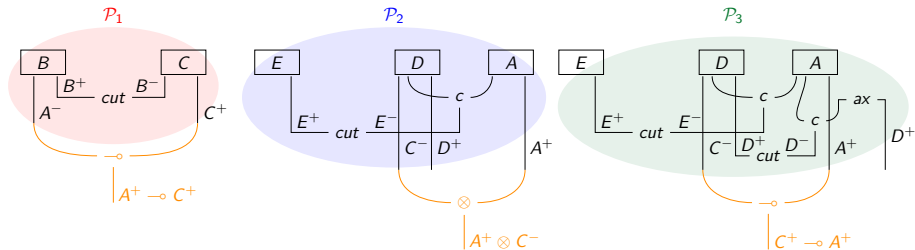
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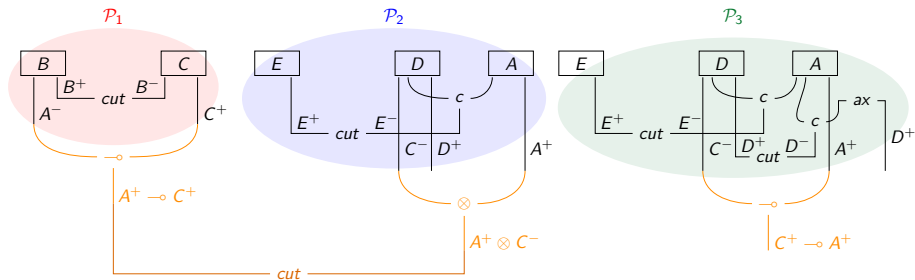
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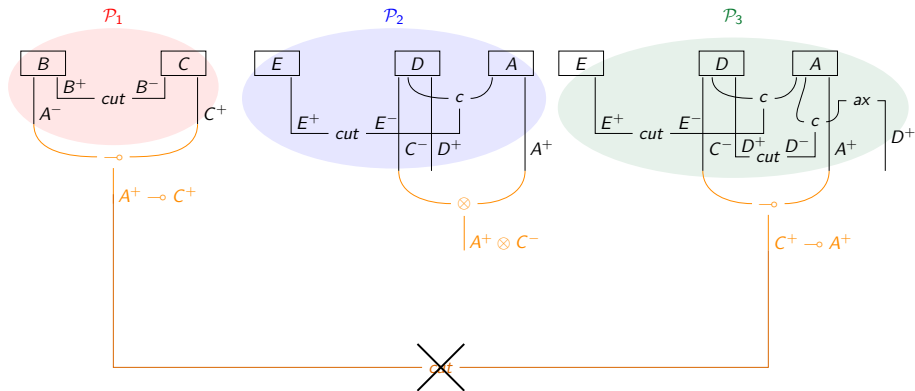
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Conclusion

Contributions

- **Compositionality** by modifying the semantic:
from quantum operations to (more general) **quantum factors**
- **Conservative** extension:
a QBN with no quantum node is **exactly** a Bayesian Network
- **Modularity** by typing in proof-nets:
parts with **dual types** are those giving a QBN

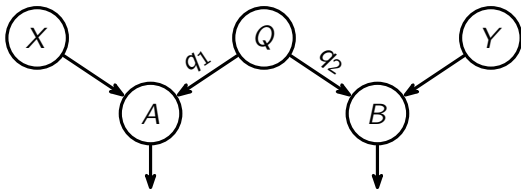
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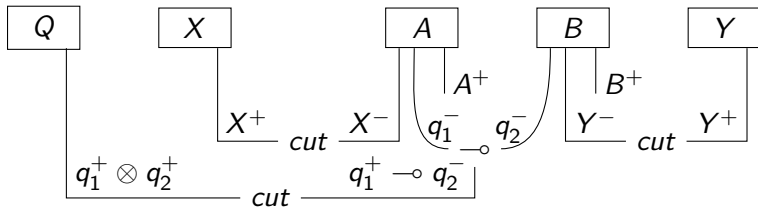
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Perspectives

- Typing using **terms** and intersection types
→ ongoing work with Claudia Faggian and Gabriele Tedeschi
- Compositionality could be used to study **conditional independence** (**no-signalling**) between random variables and/or quantum causes



Thank you for
your attention!



References



Henson, Joe, Raymond Lal, and Matthew F Pusey (Nov. 2014).
“Theory-independent limits on correlations from generalized Bayesian networks”. In: *New Journal of Physics* 16.11. ISSN: 1367-2630. DOI: 10.1088/1367-2630/16/11/113043.

Back-up: Uses of Quantum Bayesian Networks

Motivations: study probabilities of systems both quantum & classical

- **Conditional Independence / No-Signalling**

When are *classical* data independent? conditionally independent?

→ *d-separation*, a purely graphical criterion

Example: A and Y independent in Bell's experiment, but A and Y not independent given B

- **In which situations is quantum stronger than classical?**

Well-known that in Bell's Experiment quantum $>$ classical

→ DAGs where quantum $>$ classical? where quantum = classical?

